

Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

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Contents

Preface	3
1 Vector Spaces	4
1A $\mathbb{R}^n$ and $\mathbb{C}^n$	4
1B Definition of Vector Space	9
1C Subspaces	14
2 Finite-Dimensional Vector Spaces	24
2A Span and Linear Independence	24
2B Bases	33
2C Dimension	40
3 Linear Maps	51
3A Vector Space of Linear Maps	51
3B Null Spaces and Ranges	62
3C Matrices	78
3D Invertibility and Isomorphisms	88
3E Products and Quotients of Vector Spaces	99
3F Duality	108
4 Polynomials	124
5 Eigenvalues and Eigenvectors	130
5A Invariant Subspaces	130
5B The Minimal Polynomial	146
5C Upper-Triangular Matrices	162
5D Diagonalizable Operators	169
5E Commuting Operators	182
6 Inner Product Spaces	187
6A Inner Products and Norms	187
6B Orthonormal Bases	205
6C Orthogonal Complements and Minimization Problems	220

7	Operators on Inner Product Spaces	231
7A	Self-Adjoint and Normal Operators	231
7B	Spectral Theorem	246
A	Appendix: Alternate Proofs of Exercises	251
A1	Vector Spaces	251
A2	Finite-Dimensional Vector Spaces	252
A3	Linear Maps	253
A4	Polynomials	258
A5	Eigenvalues and Eigenvectors	259
A6	Inner Product Spaces	262

7 Operators on Inner Product Spaces

7A Self-Adjoint and Normal Operators

Exercise 7A1

Suppose n is a positive integer. Define $T \in \mathcal{L}(\mathbb{F}^n)$ by

$$T(z_1, \dots, z_n) = (0, z_1, \dots, z_{n-1}).$$

Find a formula for $T^*(z_1, \dots, z_n)$.

Solution. If $n = 1$, then $T(z_1) = 0$, so $T^*(z_1) = 0$. Let $n > 1$. Then for all $(z_1, \dots, z_n), (w_1, \dots, w_n) \in \mathbb{F}^n$, we have

$$\langle T(z_1, \dots, z_n), (w_1, \dots, w_n) \rangle = \langle (0, z_1, \dots, z_{n-1}), (w_1, \dots, w_n) \rangle = z_1 \overline{w_2} + \dots + z_{n-1} \overline{w_n}.$$

We need

$$\langle (z_1, \dots, z_n), T^*(w_1, \dots, w_n) \rangle = z_1 \overline{w_2} + \dots + z_{n-1} \overline{w_n}.$$

By comparison, we see that $T^*(w_1, \dots, w_n) = (w_2, \dots, w_n, 0)$, so $T^*(z_1, \dots, z_n) = (z_2, \dots, z_n, 0)$ for all $(z_1, \dots, z_n) \in \mathbb{F}^n$. ■

Exercise 7A2

Suppose $T \in \mathcal{L}(V, W)$. Prove that

$$T = 0 \iff T^* = 0 \iff T^*T = 0 \iff TT^* = 0.$$

Solution. Since $(T^*)^* = T$, then $T = 0 \iff T^* = 0$. Moreover, if $T^* = 0$, then $T^*T = 0$. Conversely, if $T^*T = 0$, then for all $v \in V$, we have

$$\|Tv\|^2 = \langle Tv, Tv \rangle = \langle v, T^*Tv \rangle = \langle v, 0 \rangle = 0.$$

Hence, $Tv = 0$ for all $v \in V$, so $T = 0$. Therefore,

$$T = 0 \iff T^* = 0 \iff T^*T = 0.$$

Replacing T with T^* in the equivalence $T = 0 \iff T^*T = 0$ gives $T^* = 0 \iff TT^* = 0$. We then have

$$T = 0 \iff T^* = 0 \iff T^*T = 0 \iff TT^* = 0. \quad \blacksquare$$

Exercise 7A3

Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. Prove that

$$\lambda \text{ is an eigenvalue of } T \iff \overline{\lambda} \text{ is an eigenvalue of } T^*.$$

Solution. Since V is finite-dimensional,

$$\dim \text{null}(T - \lambda I) = \dim V - \dim \text{range}(T - \lambda I) = \dim(\text{range}(T - \lambda I)^\perp) = \dim \text{null}((T - \lambda I)^*).$$

Hence,

$$\text{null}(T - \lambda I) \neq \{0\} \iff \text{null}((T - \lambda I)^*) \neq \{0\}.$$

Since $(T - \lambda I)^* = T^* - \bar{\lambda}I$, we have

$$\lambda \text{ is an eigenvalue of } T \iff \bar{\lambda} \text{ is an eigenvalue of } T^*. \quad \blacksquare$$

Exercise 7A4

Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V . Prove that

$$U \text{ is invariant under } T \iff U^\perp \text{ is invariant under } T^*.$$

Solution.

$$\begin{aligned} U^\perp \text{ is invariant under } T^* &\iff T^*v \in U^\perp \text{ for all } v \in U^\perp \\ &\iff \langle u, T^*v \rangle = 0 \text{ for all } u \in U, v \in U^\perp \\ &\iff \langle Tu, v \rangle = 0 \text{ for all } u \in U, v \in U^\perp \\ &\iff Tu \in (U^\perp)^\perp \text{ for all } u \in U \\ &\iff Tu \in U \text{ for all } u \in U \\ &\iff U \text{ is invariant under } T \end{aligned} \quad \blacksquare$$

Exercise 7A5

Suppose $T \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W . Prove that

$$\|Te_1\|^2 + \dots + \|Te_n\|^2 = \|T^*f_1\|^2 + \dots + \|T^*f_m\|^2.$$

Axler Note. The numbers $\|Te_1\|^2, \dots, \|Te_n\|^2$ in the equation above depend on the orthonormal basis $e_1, \dots, e_n$, but the right side of the equation does not depend on $e_1, \dots, e_n$. Thus, the equation above shows that the sum on the left side does not depend on which orthonormal basis $e_1, \dots, e_n$ is used.

Manual Remark. For students familiar with this topic already, note that the left-hand side is the squared Frobenius norm of the matrix representation of T with respect to the orthonormal bases $e_1, \dots, e_n$ and $f_1, \dots, f_m$. This exercise shows that $\sum_i \|Te_i\|^2 = \sum_j \|T^*f_j\|^2$, which is independent of the choice of orthonormal basis $e_1, \dots, e_n$. Moreover, noting that $\text{tr}(T^*T) = \text{tr}(TT^*)$ by the cyclic property of the trace, we see that the sum of the squared norms of the columns of the matrix representation of T equals the sum of the squared norms of the rows, which aligns with the result of this exercise.

Solution. Let

$$Te_j = \sum_{k=1}^m \langle Te_j, f_k \rangle f_k \implies \|Te_j\|^2 = \sum_{k=1}^m |\langle Te_j, f_k \rangle|^2.$$

Similarly, we may write

$$T^* f_i = \sum_{\ell=1}^n \langle T^* f_i, e_\ell \rangle e_\ell \implies \|T^* f_i\|^2 = \sum_{\ell=1}^n |\langle T^* f_i, e_\ell \rangle|^2.$$

Note that

$$\langle T^* f_i, e_j \rangle = \langle f_i, T e_j \rangle = \overline{\langle T e_j, f_i \rangle} \implies |\langle T^* f_i, e_j \rangle|^2 = |\langle T e_j, f_i \rangle|^2.$$

Therefore, we have

$$\sum_{k=1}^m \|T^* f_k\|^2 = \sum_{k=1}^m \sum_{\ell=1}^n |\langle T e_\ell, f_k \rangle|^2 = \sum_{\ell=1}^n \sum_{k=1}^m |\langle T e_\ell, f_k \rangle|^2 = \sum_{\ell=1}^n \|T e_\ell\|^2. \quad \blacksquare$$

Exercise 7A6

Suppose $T \in \mathcal{L}(V, W)$. Prove that

- (a) T is injective $\iff T^*$ is surjective;
- (b) T is surjective $\iff T^*$ is injective.

Solution.

- (a) T injective $\iff \text{null } T = \{0\} \iff (\text{range } T^*)^\perp = \{0\} \iff \text{range } T^* = V \iff T^*$ surjective.
- (b) Using part (a) on $T^* \in \mathcal{L}(W, V)$ and the fact that $(T^*)^* = T$, we have

$$T^* \text{ is injective} \iff (T^*)^* \text{ is surjective} \iff T \text{ is surjective.} \quad \blacksquare$$

Exercise 7A7

Prove that if $T \in \mathcal{L}(V, W)$, then

- (a) $\dim \text{null } T^* = \dim \text{null } T + \dim W - \dim V$;
- (b) $\dim \text{range } T^* = \dim \text{range } T$.

Solution.

- (a) By Theorem 7.6(a), we have $\text{null } T^* = (\text{range } T)^\perp$, so $\dim \text{null } T^* = \dim W - \dim \text{range } T$. Using the fundamental theorem of linear maps, we have $\dim \text{range } T = \dim V - \dim \text{null } T$, so

$$\dim \text{null } T^* = \dim W - \dim V + \dim \text{null } T.$$

- (b) We have $\text{range } T^* = (\text{null } T)^\perp$, so $\dim \text{range } T^* = \dim V - \dim \text{null } T = \dim \text{range } T$. ■

Exercise 7A8

Suppose A is an m -by- n matrix with entries in $\mathbb{F}$. Use (b) in Exercise 7 to prove that the row rank of A equals the column rank of A .

Axler Note. This exercise asks for yet another alternative proof of a result that was previously proved in 3.57 and 3.133.

Solution. Let $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ have the matrix $A = \mathcal{M}(T)$ with respect to the standard bases of $\mathbb{F}^n$ and $\mathbb{F}^m$. Since the standard bases of $\mathbb{F}^n$ and $\mathbb{F}^m$ are orthonormal, $A^* = \mathcal{M}(T^*)$ are the conjugate-transposed rows of A . Since complex conjugation preserves linear independence, the column rank of A^* equals the row rank of A . By Exercise 7A7(b), $\dim \text{range } T^* = \dim \text{range } T$, so the column rank of A^* equals the column rank of A , and the result follows. ■

Exercise 7A9

Prove that the product of two self-adjoint operators on V is self-adjoint if and only if the two operators commute.

Solution. Let $S, T \in \mathcal{L}(V)$ be self-adjoint, so $S^* = S$ and $T^* = T$. Note that $(ST)^* = T^*S^* = TS$. Hence,

$$ST \text{ is self-adjoint} \iff (ST)^* = ST \iff TS = ST. \quad \blacksquare$$

Exercise 7A10

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that T is self-adjoint if and only if

$$\langle Tv, v \rangle = \langle T^*v, v \rangle$$

for all $v \in V$.

Solution.

⊆ If T is self-adjoint, then $T = T^*$, so for all $v \in V$,

$$\langle Tv, v \rangle = \langle T^*v, v \rangle.$$

⊇ Suppose $\langle Tv, v \rangle = \langle T^*v, v \rangle$ for all $v \in V$. Then for all $v \in V$,

$$\langle (T - T^*)v, v \rangle = 0.$$

Since $\mathbb{F} = \mathbb{C}$, we may use 7.13 with $T - T^*$ in place of T to conclude that $T - T^* = 0$, so $T = T^*$, i.e., T is self-adjoint. ■

Exercise 7A11

Define an operator $S : \mathbb{F}^2 \rightarrow \mathbb{F}^2$ by $S(w, z) = (-z, w)$.

- (a) Find a formula for S^* .
- (b) Show that S is normal but not self-adjoint.
- (c) Find all eigenvalues of S .

Axler Note. If $\mathbb{F} = \mathbb{R}$, then S is the operator on $\mathbb{R}^2$ of counterclockwise rotation by 90° .

Solution.

- (a) For any $(w, z), (x, y) \in \mathbb{F}^2$,

$$\langle S(w, z), (x, y) \rangle = \langle (-z, w), (x, y) \rangle = -z\bar{x} + w\bar{y}.$$

Suppose $S^*(x, y) = (u, v)$. Then

$$\langle S(w, z), (x, y) \rangle = \langle (w, z), S^*(x, y) \rangle = \langle (w, z), (u, v) \rangle = w\bar{u} + z\bar{v}.$$

Comparing with $-z\bar{x} + w\bar{y}$, we get $w\bar{u} + z\bar{v} = -z\bar{x} + w\bar{y}$ for all $w, z \in \mathbb{F}$. Hence, $u = y$ and $v = -x$, so $S^*(x, y) = (y, -x)$.

(b) Observe that

$$SS^*(w, z) = S(S^*(w, z)) = S(z, -w) = (w, z),$$

and

$$S^*S(w, z) = S^*(S(w, z)) = S^*(-z, w) = (w, z).$$

Hence, $SS^* = S^*S$, so S is normal. However,

$$S(1, 0) = (0, 1) \neq (0, -1) = S^*(1, 0).$$

Hence, S is not self-adjoint.

(c) If $S(w, z) = \lambda(w, z)$ for some $\lambda \in \mathbb{F}$ and $(w, z) \neq (0, 0)$, then $-z = \lambda w$ and $w = \lambda z$. From this, we get $w = -\lambda^2 w$, so $\lambda^2 = -1$. If $\mathbb{F} = \mathbb{R}$, then S has no real eigenvalues. If $\mathbb{F} = \mathbb{C}$, then the eigenvalues of S are $\lambda = i$ and $\lambda = -i$. ■

Exercise 7A12

An operator $B \in \mathcal{L}(V)$ is called **skew** if

$$B^* = -B.$$

Suppose that $T \in \mathcal{L}(V)$. Prove that T is normal if and only if there exist commuting operators A and B such that A is self-adjoint, B is a skew operator, and $T = A + B$.

Solution.

⇐ If T is normal, then $TT^* = T^*T$. Consider the operators

$$A = \frac{T + T^*}{2}, \quad B = \frac{T - T^*}{2}.$$

Then A is self-adjoint, B is skew, and $T = A + B$. Moreover,

$$\begin{aligned} AB &= \frac{T + T^*}{2} \cdot \frac{T - T^*}{2} \\ &= \frac{1}{4}(TT - TT^* + T^*T - T^*T^*) \\ &= \frac{1}{4}(TT + TT^* - T^*T - T^*T^*) \\ &= \frac{T - T^*}{2} \cdot \frac{T + T^*}{2} = BA, \end{aligned}$$

hence A and B commute.

⇒ Suppose there exist commuting operators A and B such that A is self-adjoint, B is skew, and $T = A + B$. Then

$$T^* = (A + B)^* = A^* + B^* = A - B.$$

Hence,

$$TT^* = (A + B)(A - B) = A^2 - AB + BA - B^2 = A^2 - B^2,$$

and

$$T^*T = (A - B)(A + B) = A^2 + AB - BA - B^2 = A^2 - B^2.$$

Therefore, $TT^* = T^*T$, so T is normal. ■

Exercise 7A13

Suppose $\mathbb{F} = \mathbb{R}$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by $\mathcal{A}T = T^*$ for all $T \in \mathcal{L}(V)$.

- (a) Find all eigenvalues of $\mathcal{A}$.
- (b) Find the minimal polynomial of $\mathcal{A}$.

Solution.

- (a) If $V = \{0\}$, then $\mathcal{L}(V) = \{0\}$ and $\mathcal{A}$ has no eigenvalues. Otherwise, assume $\dim V \geq 1$. If $\mathcal{A}T = \lambda T$ for some $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{R}$, then $T^* = \lambda T$. Taking the adjoint of both sides gives

$$T = \lambda T^* = \lambda(\lambda T) = \lambda^2 T.$$

Hence, $\lambda^2 = 1$, so $\lambda = 1$ or $\lambda = -1$. If $\dim V = 1$, then $T = \lambda I$ for some $\lambda \in \mathbb{R}$. In this case, $T^* = T$, so $\mathcal{A}T = T = 1 \cdot T$, and the only eigenvalue of $\mathcal{A}$ is 1. If $\dim V \geq 2$, let $e_1, \dots, e_n$ be an orthonormal basis of V with $n = \dim V$. Define $T \in \mathcal{L}(V)$ by $Te_1 = e_2$, $Te_2 = -e_1$, and $Te_j = 0$ for $j \geq 3$. Then $T^*e_1 = -e_2$, $T^*e_2 = e_1$, and $T^*e_j = 0$ for $j \geq 3$, so $T^* = -T$. Hence, -1 is also an eigenvalue of $\mathcal{A}$. Therefore, the eigenvalues of $\mathcal{A}$ are 1 and -1 .

- (b) Observe that $\mathcal{A}(\mathcal{A}T) = \mathcal{A}T^* = (T^*)^* = T$ for all $T \in \mathcal{L}(V)$. Hence, $\mathcal{A}^2 - I = 0$, so the minimal polynomial of $\mathcal{A}$ must divide $z^2 - 1$. If $\dim V = 1$, then $\mathcal{A}$ has only one eigenvalue, 1, so the minimal polynomial of $\mathcal{A}$ is $z - 1$. Otherwise, both eigenvalues 1 and -1 occur when $\dim V \geq 2$, so the minimal polynomial of $\mathcal{A}$ is $z^2 - 1$. ■

Exercise 7A14

Define an inner product on $\mathcal{P}_2(\mathbb{R})$ by $\langle p, q \rangle = \int_0^1 pq$. Define an operator $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ by

$$T(ax^2 + bx + c) = bx.$$

- (a) Show that with this inner product, the operator T is not self-adjoint.
- (b) The matrix of T with respect to the basis $1, x, x^2$ is

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix equals its conjugate transpose, even though T is not self-adjoint. Explain why this is not a contradiction.

Solution.

- (a) Consider $p = 1$ and $q = x$. Then

$$\langle Tp, q \rangle = \langle 0, x \rangle = 0,$$

but

$$\langle p, Tq \rangle = \langle 1, x \rangle = \int_0^1 x \, dx = \frac{1}{2} \neq 0.$$

Hence, T is not self-adjoint.

(b) Observe that

$$\|x\|^2 = \langle x, x \rangle = \int_0^1 x^2 \, dx = \frac{1}{3}.$$

By 7.9, given an orthonormal basis, T is self-adjoint if and only if the matrix of T with respect to that basis equals its conjugate transpose. However, the basis $1, x, x^2$ is not orthonormal with respect to the given inner product, so the fact that the matrix of T with respect to this basis equals its conjugate transpose does not imply that T is self-adjoint. ■

Exercise 7A15

Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

- (a) T is self-adjoint $\iff T^{-1}$ is self-adjoint;
- (b) T is normal $\iff T^{-1}$ is normal.

Solution. We first note that

$$TT^{-1} = I \implies (T^{-1})^* T^* = I^* = I \implies (T^{-1})^* = (T^*)^{-1}.$$

(a)

$$\begin{aligned} T^{-1} \text{ is self-adjoint} &\iff (T^{-1})^* = T^{-1} \\ &\iff (T^*)^{-1} = T^{-1} \\ &\iff T^* = T \\ &\iff T \text{ is self-adjoint.} \end{aligned}$$

(b)

$$\begin{aligned} T^{-1} \text{ is normal} &\iff (T^{-1})^* T^{-1} = T^{-1} (T^{-1})^* \\ &\iff (T^*)^{-1} T^{-1} = T^{-1} (T^*)^{-1} \\ &\iff (T^* T)^{-1} = (T T^*)^{-1} \\ &\iff T^* T = T T^* \\ &\iff T \text{ is normal.} \end{aligned} \quad \blacksquare$$

Exercise 7A16

Suppose $\mathbb{F} = \mathbb{R}$.

- (a) Show that the set of self-adjoint operators on V is a subspace of $\mathcal{L}(V)$.
- (b) What is the dimension of the subspace of $\mathcal{L}(V)$ in (a) [in terms of $\dim V$]?]

Solution.

- (a) Let $\mathcal{S}$ be the set of self-adjoint operators on V . Since $0^* = 0$, we have $0 \in \mathcal{S}$. For $S, T \in \mathcal{S}$ and $\lambda \in \mathbb{R}$, we have

$$(S + T)^* = S^* + T^* = S + T \in \mathcal{S},$$

and

$$(\lambda T)^* = \bar{\lambda} T^* = \lambda T \in \mathcal{S}.$$

Therefore, $\mathcal{S}$ is closed under addition and scalar multiplication, and contains the zero operator, so it is a subspace of $\mathcal{L}(V)$.

- (b) Let $n = \dim V$, and let $e_1, \dots, e_n$ be an orthonormal basis of V . By 7.9, it follows that T is self-adjoint if and only if $\mathcal{M}(T)$ is a symmetric matrix, i.e., $\mathcal{M}(T) = \mathcal{M}(T)^t$. Since a symmetric $n \times n$ matrix is determined by the entries on and above the main diagonal, the dimension of the subspace of symmetric matrices is

$$\frac{n(n+1)}{2}.$$

■

Exercise 7A17

Suppose $\mathbb{F} = \mathbb{C}$. Show that the set of self-adjoint operators on V is not a subspace of $\mathcal{L}(V)$.

Solution. Note the result is not true when $V = \{0\}$, since the only operator on V is the zero operator, which is self-adjoint. Let $\mathcal{S}$ be the set of self-adjoint operators on V . If $\mathcal{S}$ were a subspace of $\mathcal{L}(V)$, consider iI , where $I \in \mathcal{S}$ is the identity operator on V . Then

$$(iI)^* = \bar{i}I = -iI \neq iI.$$

Therefore, iI is not self-adjoint, which shows that the set of self-adjoint operators on V is not a subspace of $\mathcal{L}(V)$ when $V \neq \{0\}$. ■

Exercise 7A18

Suppose $\dim V \geq 2$. Show that the set of normal operators on V is not a subspace of $\mathcal{L}(V)$.

Solution. Fix an orthonormal pair $e_1, e_2 \in V$. Extend this to an orthonormal basis $e_1, \dots, e_n$ of V . Consider the operators S and T defined by

$$Se_1 = e_2, Se_2 = e_1, Se_i = 0 \text{ for } i > 2, \quad Te_1 = -e_2, Te_2 = e_1, Te_i = 0 \text{ for } i > 2.$$

Then

$$\mathcal{M}(S) = \begin{pmatrix} 0 & 1 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

From this, we see that $\mathcal{M}(S) = \mathcal{M}(S)^t$, so S is normal. Moreover,

$$\mathcal{M}(T) = \begin{pmatrix} 0 & 1 & \cdots & 0 \\ -1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Direct computation yields $\mathcal{M}(T)\mathcal{M}(T)^* = \mathcal{M}(T)^*\mathcal{M}(T)$, so T is normal. However, $S + T$ is not normal, as can be seen by computing

$$\mathcal{M}(S + T) = \begin{pmatrix} 0 & 2 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \quad \mathcal{M}(S + T)^* = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 2 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Direct computation shows that $\mathcal{M}(S + T)\mathcal{M}(S + T)^* \neq \mathcal{M}(S + T)^*\mathcal{M}(S + T)$, so $S + T$ is not normal. ■

Exercise 7A19

Suppose $T \in \mathcal{L}(V)$ and $\|T^*v\| \leq \|Tv\|$ for every $v \in V$. Prove that T is normal.

Axler Note. This exercise fails on infinite-dimensional inner product spaces, leading to what are called [hyponormal operators](#), which have a well-developed theory.

Solution. If $\dim V = 0$, then T is trivially normal. So assume $\dim V > 0$. Note that for all $v \in V$, we have

$$\|T^*v\|^2 = \langle T^*v, T^*v \rangle = \langle TT^*v, v \rangle, \quad \|Tv\|^2 = \langle Tv, Tv \rangle = \langle T^*Tv, v \rangle.$$

From the hypothesis, we have $\|T^*v\|^2 \leq \|Tv\|^2$ for all $v \in V$, which gives

$$\langle (T^*T - TT^*)v, v \rangle \leq 0 \quad \text{for all } v \in V.$$

Let $S = TT^* - T^*T$. Then $S^* = (TT^* - T^*T)^* = TT^* - T^*T = S$, so S is self-adjoint. Moreover, $\langle Sv, v \rangle \geq 0$ for all $v \in V$. Let $e_1, \dots, e_n$ be an orthonormal basis of V . From Exercise 7A5, then

$$\sum_{i=1}^n \|Te_i\|^2 = \sum_{i=1}^n \langle T^*Te_i, e_i \rangle = \sum_{i=1}^n \langle TT^*e_i, e_i \rangle = \sum_{i=1}^n \|T^*e_i\|^2.$$

Then

$$\sum_{i=1}^n \langle Se_i, e_i \rangle = \sum_{i=1}^n \langle (TT^* - T^*T)e_i, e_i \rangle = \sum_{i=1}^n \langle TT^*e_i, e_i \rangle - \sum_{i=1}^n \langle T^*Te_i, e_i \rangle = 0.$$

Since each $\langle Se_i, e_i \rangle \geq 0$ and their sum is 0, it follows that $\langle Se_i, e_i \rangle = 0$ for all i . For arbitrary $u \in V$ with $\|u\| = 1$, we may extend it to an orthonormal basis $u, f_2, \dots, f_n$ of V . Then

$$\langle Su, u \rangle + \sum_{i=2}^n \langle Sf_i, f_i \rangle = \sum_{i=1}^n \langle Se_i, e_i \rangle = 0.$$

Since each term is nonnegative, it follows that $\langle Su, u \rangle = 0$ for all unit vectors $u \in V$. By homogeneity, $\langle Sv, v \rangle = 0$ for all $v \in V$. From 7.16, if S is self-adjoint and $\langle Sv, v \rangle = 0$ for all $v \in V$, then $S = 0$. Hence, $TT^* - T^*T = 0$, so $TT^* = T^*T$, and therefore T is normal. ■

Exercise 7A20

Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that the following are equivalent.

- (a) P is self-adjoint.
- (b) P is normal.
- (c) There is a subspace U of V such that $P = P_U$.

Solution.

(a) $\implies$ (b): If P is self-adjoint, then $P = P^*$. Hence,

$$PP^* = PP = P^2 = P = P^*P = P^*P,$$

so P is normal.

(b) $\implies$ (c): Since P is normal,

$$(\text{range } P)^\perp = \text{null } P^* = \text{null } P.$$

Since $P^2 = P$, $V = \text{null } P \oplus \text{range } P$ is an orthogonal direct sum. Letting $U = \text{range } P$, then any $v \in V$ can be written uniquely as $v = u + w$ with $u \in U$ and $w \in \text{null } P$. Hence,

$$Pv = P(u + w) = Pu + Pw = u + 0 = u \in U.$$

Therefore, $P = P_U$.

(c) $\implies$ (a): Write $V = U \oplus U^\perp$. Then for $v = u_1 + w_1$ and $w = u_2 + w_2$ with $u_1, u_2 \in U$ and $w_1, w_2 \in U^\perp$, we have

$$\langle P_U v, w \rangle = \langle u_1, u_2 + w_2 \rangle = \langle u_1, u_2 \rangle = \langle u_1 + w_1, u_2 \rangle = \langle v, P_U w \rangle.$$

Hence, P_U is self-adjoint. ■

Exercise 7A21

Suppose $D : \mathcal{P}_8(\mathbb{R}) \rightarrow \mathcal{P}_8(\mathbb{R})$ is the differentiation operator defined by $Dp = p'$. Prove that there does not exist an inner product on $\mathcal{P}_8(\mathbb{R})$ that makes D a normal operator.

Solution. If there did exist an inner product on $\mathcal{P}_8(\mathbb{R})$ that makes D a normal operator, then $\|Dp\| = \|D^*p\|$ for all $p \in \mathcal{P}_8(\mathbb{R})$, by 7.20. Consider $p = x$ so that $Dp = 1 \neq 0$, but $D^2p = 0$. Then applying 7.20 to Dp , we get

$$\|D^*(Dp)\| = \|D(Dp)\| = \|D^2p\| = \|0\| = 0$$

so that $D^*(Dp) = 0$. Then

$$\langle D^*(Dp), p \rangle = \langle Dp, Dp \rangle = \|Dp\|^2.$$

Since $D^*(Dp) = 0$, the left-hand side of the above equation is 0, while the right-hand side is $\|Dp\|^2 \neq 0$. This is a contradiction, so no such inner product can exist. ■

Exercise 7A22

Give an example of an operator $T \in \mathcal{L}(\mathbb{R}^3)$ such that T is normal but not self-adjoint.

Solution. Let T have the matrix

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

with respect to the standard basis of $\mathbb{R}^3$, which is orthonormal. Then T is normal because

$$T^* = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathcal{M}(T)^*.$$

By direct computation, we have

$$TT^* = I, \quad T^*T = I.$$

Hence, T is normal but not self-adjoint, since $T \neq T^*$. ■

Exercise 7A23

Suppose T is a normal operator on V . Suppose also that $v, w \in V$ satisfy the equations

$$\|v\| = \|w\| = 2, \quad Tv = 3v, \quad Tw = 4w.$$

Show that $\|T(v + w)\| = 10$.

Solution. Since v, w are eigenvectors of the normal operator T corresponding to distinct eigenvalues 3 and 4, they must be orthogonal. Therefore, $\langle v, w \rangle = 0$. Then

$$\begin{aligned} \|T(v + w)\|^2 &= \langle Tv + Tw, Tv + Tw \rangle \\ &= \langle 3v + 4w, 3v + 4w \rangle \\ &= \langle 3v, 3v \rangle + \langle 3v, 4w \rangle + \langle 4w, 3v \rangle + \langle 4w, 4w \rangle \\ &= 9\langle v, v \rangle + 12\langle v, w \rangle + 12\langle w, v \rangle + 16\langle w, w \rangle \\ &= 9\|v\|^2 + 16\|w\|^2 \\ &= 9 \cdot 4 + 16 \cdot 4 \\ &= 100. \end{aligned}$$

Therefore, $\|T(v + w)\| = 10$. ■

Exercise 7A24

Suppose $T \in \mathcal{L}(V)$ and

$$a_0 + a_1z + a_2z^2 + \cdots + a_{m-1}z^{m-1} + z^m$$

is the minimal polynomial of T . Prove that the minimal polynomial of T^* is

$$\overline{a_0} + \overline{a_1}z + \overline{a_2}z^2 + \cdots + \overline{a_{m-1}}z^{m-1} + z^m.$$

Axler Note. This exercise shows that the minimal polynomial of T^* equals the minimal polynomial of T if $\mathbb{F} = \mathbb{R}$.

Solution. Define $p(z)$ to be the minimal polynomial of T . Let $q(z) = \sum_{i=0}^m \overline{a_i}z^i$. Then

$$q(T^*) = \sum_{i=0}^m \overline{a_i}(T^*)^i = \sum_{i=0}^m \overline{a_i}(T^i)^* = \left(\sum_{i=0}^m a_i T^i \right)^* = p(T)^* = 0.$$

Hence, $q(z)$ is a polynomial that annihilates T^* . Using a similar argument with $r(z)$ being the minimal polynomial of T^* , we can show that $\bar{r}(T) = 0$. Since $q(z)$ annihilates T^* and $r(z)$ is the minimal polynomial of T^* , it follows that $r(z)$ divides $q(z)$. Similarly, since $\bar{r}(T) = 0$, $p(z)$ divides $\bar{r}(z)$. Then $\deg r \leq \deg q$ from the fact that $r(z)$ divides $q(z)$, and $\deg q \leq \deg r$ from the fact that $p(z)$ divides $\bar{r}(z)$ and $\deg p = \deg q$. Hence, $\deg r = \deg q$, and since $r(z)$ is monic and divides $q(z)$, we must have $r(z) = q(z)$. Therefore, the minimal polynomial of T^* is $q(z)$. ■

Exercise 7A25

Suppose $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if T^* is diagonalizable.

Solution.

$\Rightarrow$ Suppose T is diagonalizable. Then the minimal polynomial of T splits into distinct linear factors by 5.62. By Exercise 7A24, the minimal polynomial of T^* also splits into distinct linear factors whose roots are the complex conjugates of the roots of the minimal polynomial of T . Hence, T^* is diagonalizable.

$\Leftarrow$ Conversely, suppose T^* is diagonalizable. Apply the same argument as above, using the fact that $(T^*)^* = T$. Therefore, T is diagonalizable. ■

Exercise 7A26

Fix $u, x \in V$. Define $T \in \mathcal{L}(V)$ by $Tv = \langle v, u \rangle x$ for every $v \in V$.

- (a) Prove that if V is a real vector space, then T is self-adjoint if and only if the list u, x is linearly dependent.
- (b) Prove that T is normal if and only if the list u, x is linearly dependent.

Solution. Note that

$$\langle Tv, w \rangle = \langle \langle v, u \rangle x, w \rangle = \langle v, u \rangle \langle x, w \rangle = \langle v, \overline{\langle x, w \rangle} u \rangle = \langle v, T^*w \rangle.$$

Since $\langle v, u_1 \rangle = \langle v, u_2 \rangle$ for all $v \in V$ implies $u_1 = u_2$, we conclude that $T^*w = \overline{\langle x, w \rangle} u = \langle w, x \rangle u$.

- (a) $\Rightarrow$ If T is self-adjoint, then $T = T^*$, which means $\langle v, u \rangle x = \langle v, x \rangle u$ for all $v \in V$. If $u = 0$ or $x = 0$, then the list u, x is linearly dependent. If both are nonzero, then letting $v = u$ gives $\|u\|^2 x = \langle u, x \rangle u$, hence

$$x = \frac{\langle u, x \rangle}{\|u\|^2} u.$$

Hence, x and u are linearly dependent.

$\Leftarrow$ Conversely, suppose one of u or x is zero. Then $T = 0 = T^*$, so T is self-adjoint. If both are nonzero and linearly dependent, let $x = \lambda u$ for some $\lambda \in \mathbb{R}$. Then for all $v \in V$,

$$Tv = \langle v, u \rangle x = \langle v, u \rangle \lambda u = \lambda \langle v, u \rangle u = T^*v,$$

hence T is self-adjoint.

- (b) $\Rightarrow$ Suppose T is normal. Then for any $y \in V$,

$$TT^*y = T(\langle y, x \rangle u) = \langle y, x \rangle Tu = \langle y, x \rangle \langle u, u \rangle x = \|u\|^2 \langle y, x \rangle x,$$

and

$$T^*Ty = T^*(\langle y, u \rangle x) = \langle y, u \rangle T^*x = \langle y, u \rangle \langle x, x \rangle u = \|x\|^2 \langle y, u \rangle u.$$

If $u = 0$ or $x = 0$, then the list u, x is linearly dependent. Otherwise, for nonzero u and x , we must have $\|u\|^2 \langle y, x \rangle x = \|x\|^2 \langle y, u \rangle u$ for all $y \in V$. Let $y = u$. Then $\|u\|^2 \langle u, x \rangle x = \|x\|^2 \|u\|^2 u$, hence $\langle u, x \rangle x = \|x\|^2 u$. If $x \neq 0$, then

$$u = \frac{\langle u, x \rangle}{\|x\|^2} x,$$

so u and x are linearly dependent.

◁ If one of u or x is zero, then $T = 0 = T^*$, so T is normal. If both are nonzero and linearly dependent, let $u = \mu x$ for some $\mu \in \mathbb{F}$. Then for all $y \in V$,

$$TT^*y = \|u\|^2 \langle y, x \rangle x = \|\mu x\|^2 \langle y, x \rangle x = |\mu|^2 \|x\|^2 \langle y, x \rangle x,$$

and

$$T^*Ty = \|x\|^2 \langle y, u \rangle u = \|x\|^2 \langle y, \mu x \rangle \mu x = |\mu|^2 \|x\|^2 \langle y, x \rangle x.$$

Hence, $TT^*y = T^*Ty$ for all $y \in V$, so T is normal. ■

Exercise 7A27

Suppose $T \in \mathcal{L}(V)$ is normal. Prove that

$$\text{null } T^k = \text{null } T \quad \text{and} \quad \text{range } T^k = \text{range } T$$

for every positive integer k .

Solution. Let $v \in \text{null } T$. Then $T^k v = 0$, so $v \in \text{null } T^k$. Hence, $\text{null } T \subseteq \text{null } T^k$. Conversely, let $v \in \text{null } T^k$. Since T is normal, $\|Tv\| = \|T^*v\|$ by 7.20. Hence, $\text{null } T = \text{null } T^*$. If $T^2 v = 0$, then $T(Tv) = 0$, so $Tv \in \text{null } T = \text{null } T^*$. Therefore, $T^*Tv = 0$ so that $\|Tv\|^2 = \langle Tv, Tv \rangle = \langle T^*Tv, v \rangle = 0$. Hence, $Tv = 0$, which shows that $v \in \text{null } T$. By induction on k , assume that $\text{null } T^{k-1} = \text{null } T$. If $v \in \text{null } T^k$, then $Tv \in \text{null } T^{k-1} = \text{null } T$, so $T(Tv) = T^2v = 0$. By the base case earlier, $v \in \text{null } T^2 = \text{null } T$. Hence, by induction, $\text{null } T^k = \text{null } T$ for all positive integers k .

For the other equality, note that $v \in \text{range } T^k$ implies that there exists $u \in V$ such that $v = T^k u$. Then $T(T^{k-1}u) = T^k u = v$, so $v \in \text{range } T$. Hence, $\text{range } T^k \subseteq \text{range } T$. Conversely, note that

$$\dim \text{range } T^k = \dim V - \dim \text{null } T^k = \dim V - \dim \text{null } T = \dim \text{range } T$$

by the previous equality $\text{null } T^k = \text{null } T$. Since $\text{range } T^k \subseteq \text{range } T$ and they have the same dimension, we conclude that $\text{range } T^k = \text{range } T$. ■

Exercise 7A28

Suppose $T \in \mathcal{L}(V)$ is normal. Prove that if $\lambda \in \mathbb{F}$, then the minimal polynomial of T is not a polynomial multiple of $(x - \lambda)^2$.

Solution. If the minimal polynomial $p(x)$ of T is a polynomial multiple of $(x - \lambda)^2$, then $p(x) = (x - \lambda)^2 q(x)$ for some polynomial $q(x)$. Then $(T - \lambda I)^2 q(T) = 0$, but $(T - \lambda I)q(T) \neq 0$ since $p(x)$ is the minimal polynomial. Pick v such that $(T - \lambda I)q(T)v \neq 0$. Then $(T - \lambda I)^2 q(T)v = 0$, so $q(T)v \in \text{null}(T - \lambda I)^2$. By normality of T , we know that $(T - \lambda I)^* = T^* - \bar{\lambda}I$ which commutes with $T - \lambda I$. Hence, by Exercise 7A27, $\text{null}(T - \lambda I)^2 = \text{null}(T - \lambda I)$. Therefore, $q(T)v \in \text{null}(T - \lambda I)$, which implies that $(T - \lambda I)q(T)v = 0$. This contradicts our choice of v . Hence, the minimal polynomial of T cannot be a polynomial multiple of $(x - \lambda)^2$. ■

Exercise 7A29

Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and there is an orthonormal basis $e_1, \dots, e_n$ of V such that $\|Te_k\| = \|T^*e_k\|$ for each $k = 1, \dots, n$, then T is normal.

Solution. Let $V = \mathbb{F}^2$, and consider the standard orthonormal basis e_1, e_2 . Define $T \in \mathcal{L}(V)$ by $T(e_1) = e_1 - e_2$ and $T(e_2) = e_1$. Then

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathcal{M}(T^*) = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$$

so that $T^*e_1 = e_1 + e_2$ and $T^*e_2 = -e_1$. Moreover,

$$\|Te_1\|^2 = \|e_1 - e_2\|^2 = 2, \quad \|T^*e_1\|^2 = \|e_1 + e_2\|^2 = 2,$$

and

$$\|Te_2\|^2 = \|e_1\|^2 = 1, \quad \|T^*e_2\|^2 = \|-e_1\|^2 = 1.$$

Hence, $\|Te_k\| = \|T^*e_k\|$ for each $k = 1, 2$. However, T is not normal because

$$\mathcal{M}(TT^*) = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, \quad \mathcal{M}(T^*T) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Therefore, $TT^* \neq T^*T$. ■

Exercise 7A30

Suppose that $T \in \mathcal{L}(\mathbb{F}^3)$ is normal and $T(1, 1, 1) = (2, 2, 2)$. Suppose $(z_1, z_2, z_3) \in \text{null } T$. Prove that $z_1 + z_2 + z_3 = 0$.

Solution. Since T is normal, we have $\text{null } T = \text{null } T^*$ by 7.20. Let $z = (z_1, z_2, z_3) \in \text{null } T = \text{null } T^*$. By $T(1, 1, 1) = 2(1, 1, 1)$ and the definition of the adjoint, we have

$$2\langle(1, 1, 1), z\rangle = \langle T(1, 1, 1), z\rangle = \langle(1, 1, 1), T^*z\rangle = \langle(1, 1, 1), 0\rangle = 0.$$

Then $\langle(1, 1, 1), z\rangle = 0$. Direct calculation shows that

$$\langle(1, 1, 1), z\rangle = \overline{z_1} + \overline{z_2} + \overline{z_3} = \overline{z_1 + z_2 + z_3}.$$

Since $\langle(1, 1, 1), z\rangle = 0$, we have $\overline{z_1 + z_2 + z_3} = 0$, which implies $z_1 + z_2 + z_3 = 0$. ■

Exercise 7A31

Fix a positive integer n . In the inner product space of continuous real-valued functions on $[-\pi, \pi]$ with inner product $\langle f, g \rangle = \int_{-\pi}^{\pi} fg$, let

$$V = \text{span}(1, \cos x, \cos 2x, \dots, \cos nx, \sin x, \sin 2x, \dots, \sin nx).$$

- Define $D \in \mathcal{L}(V)$ by $Df = f'$. Show that $D^* = -D$. Conclude that D is normal but not self-adjoint.
- Define $T \in \mathcal{L}(V)$ by $Tf = f''$. Show that T is self-adjoint.

Solution.

- (a) By a similar set of arguments as in Exercise 6B4, the list above is orthogonal with respect to the given inner product. Note that any $f, g \in V$ is a linear combination of $1, \cos kx, \sin kx$ for $k = 1, 2, \dots, n$, all of which are periodic with period 2π . Then $f(\pi) = f(-\pi)$ and $g(\pi) = g(-\pi)$. Integration by parts gives

$$\langle Df, g \rangle = \int_{-\pi}^{\pi} f'g = fg \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} fg' = - \int_{-\pi}^{\pi} fg' = \langle f, -Dg \rangle.$$

Then $D^* = -D$, which shows that

$$DD^* = D(-D) = -D^2 \quad \text{and} \quad D^*D = (-D)D = -D^2.$$

Hence, $DD^* = D^*D$, which shows that D is normal. However, D is not self-adjoint, since $D \cos x = -\sin x \neq \sin x = D^* \cos x$.

- (b) Note that $T = D^2$. By the previous part, $D^* = -D$, so $T^* = (D^2)^* = (D^*)^2 = (-D)^2 = D^2 = T$. Hence, T is self-adjoint. ■

Exercise 7A32

Suppose $T : V \rightarrow W$ is a linear map. Show that under the standard identification of V with V' (see 6.58) and the corresponding identification of W with W' , the adjoint map $T^* : W \rightarrow V$ corresponds to the dual map $T' : W' \rightarrow V'$. More precisely, show that

$$T'(\varphi_w) = \varphi_{T^*w}$$

for all $w \in W$, where φ_w and φ_{T^*w} are defined as in 6.58.

Solution. For all $v \in V$ and $w \in W$, we have

$$(T'(\varphi_w))v = \varphi_w(Tv) = \langle Tv, w \rangle = \langle v, T^*w \rangle = \varphi_{T^*w}v.$$

Hence, $T'(\varphi_w) = \varphi_{T^*w}$ for all $w \in W$. ■